

Commerce XI Maths App

Activities

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 3 - Skewness | Total questions: 12

Question 1 - ENTER ANSWER · PG-44 · A3.1

For the representative marks of 12 students arranged in ascending order:

40, 45, 48, 50, 52, 55, 58, 60, 62, 65, 68, 72

Find Q_1 .

Correct answer: 48.5

Explanation and steps

The Q_1 position is $\frac{13}{4} = 3.25$. Interpolating between the 3rd and 4th observations gives 48.5.

Question 2 - ENTER ANSWER · PG-44 · A3.1

For the representative marks:

40, 45, 48, 50, 52, 55, 58, 60, 62, 65, 68, 72

Find Q_2 .

Correct answer: 56.5

Explanation and steps

The Q_2 position is $\frac{13}{2} = 6.5$, midway between 55 and 58, so $Q_2 = 56.5$.

Question 3 - ENTER ANSWER · PG-44 · A3.1

For the representative marks:

40, 45, 48, 50, 52, 55, 58, 60, 62, 65, 68, 72

Find Q_3 .

Correct answer: 64.25

Explanation and steps

The Q_3 position is $\frac{39}{4} = 9.75$. Interpolating between 62 and 65 gives 64.25.

Question 4 - ENTER ANSWER · PG-44 · A3.1

For the representative marks:

40, 45, 48, 50, 52, 55, 58, 60, 62, 65, 68, 72

Calculate Bowley's coefficient of skewness correct to four decimal places.

Correct answer: -0.0159

Explanation and steps

$$S_{kb} = \frac{64.25 + 48.5 - 2 \times 56.5}{64.25 - 48.5} = \frac{-0.25}{15.75} \approx -0.0159.$$

Question 5 - MCQ · PG-44 · A3.1

For the representative marks:

40, 45, 48, 50, 52, 55, 58, 60, 62, 65, 68, 72

The distribution is

Options

A. slightly negatively skewed

B. positively skewed

C. perfectly symmetric

D. strongly positively skewed

Correct answer: slightly negatively skewed

Explanation and steps

Skb is slightly below zero.

Question 6 - MCQ · PG-44 · A3.1

In Activity 3.1, a different class data set may produce

Options

A. a different coefficient and comment

B. the same quartiles always

C. $Sk_b = 1$ always

D. no median

Correct answer: a different coefficient and comment

Explanation and steps

The textbook activity asks students to collect their own data, so results depend on the collected marks.

Question 7 - MCQ · PG-44 · A3.2

x	10	20	30	40	50	60	70	80	90	100
f	3	9	16	14	13	10	8	5	5	2

The largest frequency occurs at x =

Options

A. 30

B. 20

C. 40

D. 50

Correct answer: 30

Explanation and steps

The maximum frequency 16 occurs at x = 30.

Question 8 - MCQ · PG-44 · A3.2

x	10	20	30	40	50	60	70	80	90	100
f	3	9	16	14	13	10	8	5	5	2

The freehand curve has a longer tail towards

Options

A. higher x-values

B. lower x-values

C. both sides equally

D. the y-axis

Correct answer: higher x-values

Explanation and steps

Frequencies decline gradually from the peak towards 100.

Question 9 - MCQ · PG-44 · A3.2

x	10	20	30	40	50	60	70	80	90	100
f	3	9	16	14	13	10	8	5	5	2

The distribution is

Options

- A. positively skewed
- B. negatively skewed
- C. symmetric
- D. without a mode

Correct answer: positively skewed

Explanation and steps

Its longer tail extends to the right.

Question 10 - MCQ · PG-44 · A3.2

x	10	20	30	40	50	60	70	80	90	100
f	3	9	16	14	13	10	8	5	5	2

For this distribution, the expected relation is

Options

- A. Mean > Median > Mode
- B. Mode > Median > Mean
- C. Mean = Median = Mode
- D. Median > Mode > Mean

Correct answer: Mean > Median > Mode

Explanation and steps

Positive skewness gives Mean > Median > Mode.

Question 11 - MCQ · PG-44 · A3.2

x	10	20	30	40	50	60	70	80	90	100
f	3	9	16	14	13	10	8	5	5	2

The mode is

Options

- A. 30
- B. 40
- C. 50
- D. 100

Correct answer: 30

Explanation and steps

The modal x-value is the one with maximum frequency.

Question 12 - MCQ · PG-44 · A3.2

x	10	20	30	40	50	60	70	80	90	100
f	3	9	16	14	13	10	8	5	5	2

The total frequency is

Options

- A. 85
- B. 75
- C. 90
- D. 100

Correct answer: 85

Explanation and steps

Adding all frequencies gives 85.